

ALBERT KABANDA

PhD Candidate | Seismologist / Geophysicist | Ambient Noise Imaging | Receiver Functions

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PROFILE

Seismologist/geophysicist and PhD candidate studying continental rifting in the northern Western Branch of the East African Rift System. I use ambient-noise interferometry and tomography, receiver functions, waveform inversion, and earthquake waveform analysis to estimate sediment thickness, seismic velocity structure, Moho depth, and lateral crustal variations along and across the rift. I develop Python/ObsPy workflows for continuous waveform processing and large station-pair analyses on HPC systems, with additional experience in broadband seismic field operations, laboratory instruction, and technical reporting.

CORE SKILLS

- Seismic imaging: ambient-noise cross-correlation and tomography; surface-wave phase and group velocity analysis; receiver functions; H- κ stacking; waveform inversion; and CCP imaging.
- Subsurface interpretation: I have imaged sediment thickness, crustal discontinuities, Moho geometry, one-dimensional velocity structure, and lateral low- and high-velocity anomalies along and across the northern Western Branch of the East African Rift System including the unrifted zones.
- Scientific computing: I use Python, ObsPy, NumPy, SciPy, pandas, scikit-learn, Matplotlib, PyGMT, Bash/Unix, Git/GitHub, Jupyter, SAC, and SLURM-based HPC workflows.
- Field and seismic data systems: broadband network installation and maintenance, MiniSEED and StationXML preparation, continuous and event data QC, metadata, EarthScope archiving, and remote field logistics.

RESEARCH AND PROFESSIONAL EXPERIENCE

PhD Research | Northwestern University, Evanston, IL: **Jun 2022–Present**

- Conduct seismic imaging of the northern Western Branch of the East African Rift System and the Kivu–Virunga region using continuous ambient-noise records and earthquakes.
- Developed Python/ObsPy workflows that index continuous MiniSEED, identify station overlap, perform quality control and preprocessing, compute daily cross-correlations, build stacks, and prepare dispersion measurements for large station-pair datasets.
- Run multi-station, multi-year processing workflows on cluster/HPC systems and document processing decisions for reproducible analysis.
- Integrate receiver functions, H- κ stacking, waveform inversion, and related earthquake waveform analyses to constrain crustal structure and velocity models across Uganda.
- Crustal structure study of Greenland (ongoing study)

Field Seismologist | DRIAR Broadband Seismic Experiment, Uganda: **Jun 2022-Aug. 2024**

- Installed, serviced, decommissioned, and quality-controlled a 20-station broadband seismic network across remote areas of western Uganda using STS-2 and Trillium 120P sensors.
- Managed site selection, vault construction, sensor orientation and leveling, digitizers, GPS, solar and battery systems, cabling, equipment inventories, troubleshooting, and field logistics.

- Used PASSCAL tools for data QC, converted raw data to MiniSEED, created StationXML metadata, checked event and continuous records, and prepared data for EarthScope archiving.

Field Geophysicist | Geological Survey and Mines, Uganda

Jul 2014–Sep 2015

- Supported seismic, electrical resistivity, induced polarization, electromagnetic, gravity, and magnetic field surveys; processed and presented geophysical data and worked with local earthquake records in SEISAN.

Master's student | University of Bergen – Bergen, Norway

Aug 2012–Jun 2014

- Developed rock-physics templates for reservoir characterization and 4-D seismic reservoir monitoring and tested how elastic properties respond to changes in reservoir conditions.

ADDITIONAL EXPERIENCE

Sep 2016–May 2022

Physics Laboratory Specialist | Petroleum Institute and Khalifa University, Abu Dhabi, UAE

- Delivered physics laboratory demonstrations and taught undergraduate laboratory sessions, including the setup and proper use of experimental equipment. Trained junior faculty and instructors in the operation and instructional use of physics laboratory equipment.

Assistant Lecturer, Geophysics | Nkumba University

Jul 2014–Aug 2016

Entebbe, Uganda

- Taught petroleum and reservoir geophysics, reservoir characterization, thermodynamics, and petroleum geology.
- Advised student projects and presented technical material to undergraduate audiences.

EDUCATION AND SPECIALIZED TRAINING

- PhD, Earth and Planetary Sciences (Geophysics/Seismology), Northwestern University, GPA 3.83/4.0; Advisor: Prof. Suzan van der Lee

Expected Jun 2027

- MS, Earth and Planetary Sciences (Geophysics/Seismology), Northwestern University, GPA 3.83/4.0

Jun 2024

- Integrated Data Science Certificate, Northwestern University

Mar 2026

- MS, Geosciences (Reservoir Geophysics), University of Bergen

Jun 2014

- BS, Physics; Minor in Mathematics, Makerere University

Jun 2010

- **Blake Plateau/Carolina Trough Active-Source Seismic Investigation Workshop, UTIG, Jan 2026:** marine reflection/refraction, travel-time picking, layered P-wave velocity modeling, ray tracing, and model–data comparison.

SELECTED SEISMIC PROJECTS

- Seismic ambient-noise tomography, Uganda (in progress): measure phase and group velocity from station-pair correlations, develop dispersion datasets, and perform inversion and resolution testing to interpret lateral velocity variations with rift geology.
- Directionality and seasonality of ambient seismic noise and implications for subsurface imaging (submitted): test daily, monthly, seasonal, and full-duration stacks; evaluate positive and negative branches across primary and secondary microseism bands and assess temporal stability before tomography or velocity-change analysis.

- Receiver-function study, Uganda (resubmitted): used a 53-station dataset, H- κ stacking, waveform inversion, and CCP imaging to estimate sediment thickness, Moho depth, $\kappa = \alpha/\beta$, and one-dimensional shear-wave velocity structure.
- Kivu–Virunga receiver-function study (submitted): compare crustal thickness, κ , shallow sediment/volcanic effects, and one-dimensional velocity models across volcanic centers, rift basins, and uplifted flanks.
- Investigate directionality, seasonality and crustal structure variations in Greenland (in progress). I have been developing an automated seismic-noise processing workflow for Greenland that downloads and preprocesses continuous waveform data, computes daily cross-correlations for all station pairs, and prepares the resulting correlation functions for ambient-noise analysis and temporal velocity-change (dv/v) monitoring.

PUBLICATIONS

1. **Kabanda, A.**, van der Lee, S., Atekwana, E. A., Stamps, D. S., Alonzo, B., Atekwana, E. A., et al. (2026). Receiver function imaging of Uganda's crust and its magma-assisted to magma-poor rift transition in the Northern Western Branch of the East African Rift System. *Manuscript resubmitted to Journal of Geophysical Research: Solid Earth*.
2. **Kabanda, A.**, van der Lee, S., Gensini, V., Atekwana, E. A., Stamps, D. S., Atekwana, E. A., et al. (2026). Directionality and seasonality of ambient seismic noise from Africa's circumfluent oceans and its implications for subsurface imaging. *Manuscript submitted to Geophysical Journal International*.
3. **Kabanda, A.**, Kalugana, C., van der Lee, S., Ebinger, C. J., Kyambikwa, A., & Mavonga, G. T. (2026). Crustal structure beneath the Kivu Rift and Virunga Volcanic Province, Western Rift, Africa. *Manuscript submitted to Geophysical Journal International*.
4. Mwongyera, H., Taylor, M. H., Stamps, D. S., Atekwana, E. A., Evans, R. L., Barry, P. H., van der Lee, S., Katumwehe, A., Njinju, E., **Kabanda, A.**, Kwagalakwe, A., Kiberu, J. M., Nakajigo, J., & Tugume, F. (2026). Active fault database for the Northern Western Branch of the East African Rift System. *GSA Today*, 36(8), 4–11. <https://doi.org/10.1130/GSATG634A.1>
5. Kwagalakwe, A., Stamps, D. S., Kolawole, F., Atekwana, E. A., Taylor, M., Atekwana, E. A., Katumwehe, A. B., Barry, P. H., Njinju, E. A., Nyago, J., Tugume, F., Kiberu, J. M., Nakajigo, J., **Kabanda, A.**, Kianji, G., & Saria, E. (2025). Constraining the kinematics of the Victoria Microplate and the Northern Western Branch of the East African Rift System. *Geophysical Research Letters*, 52(20), e2025GL116301. <https://doi.org/10.1029/2025GL116301>
6. Njinju, E. A., Kolawole, F., Atekwana, E. A., Ajala, R., Atekwana, E. A., Stamps, D. S., Evans, R. L., Katumwehe, A., Barry, P. H., van der Lee, S., Kiberu, J. M., Tugume, F., **Kabanda, A.**, Taylor, M., Nakajigo, J., & Isaac, A. O. (2025). Density perturbations in the crust indicate potential for blind magmatism beneath magma-poor rifts. *Tectonophysics*, 913, 230881. <https://doi.org/10.1016/j.tecto.2025.230881>
7. **Kabanda, A. (2017)**. Rock physics template (RPT) technology for reservoir characterization and 4D seismic monitoring. *CSEG Recorder*, 42(1).

CONFERENCE PRESENTATIONS AND WORKSHOPS

2026

Kabanda, A., Babirye, G. P., Nseko, Y., Nyago, J., Schirbel, L., Tugume, F., van der Lee, S., Atekwana, E. A., Stamps, D. S., Atekwana, E. A., Barry, P. H., Evans, R. L., Katumwehe, A. B., Kiberu, J. M., Kolawole, F., Kwagalakwe, A., Mwongyera, H., Nakajigo, J., Njinju, E. A., Rhodes, R., Taylor, M. H., & Xue, L. (2026). *Inferring lithospheric structure along the northern Western Branch of the East African*

Rift System from seismic array analyses. Abstract submitted for presentation at the AGU Annual Meeting 2026, San Francisco, California.

Kabanda, A. (2026). *Directionality and seasonality in seismic ambient noise from circumfluent oceans of Africa, and its implications for tomographic imaging*. Presentation at the Purdue PCAG Workshop, Purdue University, May 2026.

Kabanda, A., & van der Lee, S. (2026). *Monitoring seasonality and directionality in ambient seismic noise generated by Africa's circumfluent oceans*. Oral presentation at the Seismological Society of America Annual Meeting, Pasadena, California, April 15, 2026. **SSA Student Travel Grant recipient.**

Kabanda, A., & van der Lee, S. (2026). Continental Rifting and Rifted Margins Workshop, University of Texas Institute for Geophysics (UTIG), January 12–16, 2026. Presented research and participated in interdisciplinary discussions on continental-rift structure, magmatism, sedimentary basins, and geophysical imaging.

2025

Kabanda, A., van der Lee, S., Stamps, D. S., Atekwana, E. A., Evans, R. L., Taylor, M. H., Katumwehe, A. B., Atekwana, E. A., Njinju, E. A., Tugume, F., Kiberu, J. M., Xue, L., Barry, P. H., Kolawole, F., Kwagalakwe, A., Mwongyera, H., Mongovin, D., Nakajigo, J., Nyago, J., Kabenge, L., Nagudi, B., Abbey, O. I., Yip, H., Fishwick, S., Halldorsson, S. A., & Rümpker, G. (2025). *Crustal structure around the northern Western Branch of the East African Rift System from receiver function analysis*. Oral presentation at the AGU Fall Meeting, New Orleans, Louisiana.

Kabanda, A., van der Lee, S., Tugume, F., Nyago, J., Kabenge, L., Babirye, G. P., Nseko, Y., & Alonzo, B. (2025). *Wildlife, illness, equipment malfunction, flooding, and adaptive collaboration during the DRIAR broadband seismic experiment in Uganda*. Oral presentation at the AGU Fall Meeting, New Orleans, Louisiana.

2024

Kabanda, A., van der Lee, S., Stamps, D. S., Atekwana, E. A., Nyago, J., Tugume, F., Kabenge, L., Babirye, G. P., Fishwick, S., Kolawole, F., Evans, R. L., Taylor, M. H., Katumwehe, A. B., Atekwana, E. A., & Kiberu, J. M. (2024). *Crustal structure around the northern Western Branch of the East African Rift System from receiver function analysis*. Oral presentation at the AGU Fall Meeting.

Kabanda, A. (2024). *Crustal structure of the East African Rift System in Uganda from receiver function analysis*. Lightning talk, Northwestern Institute on Complex Systems (NICO), Northwestern University, November 20, 2024.

Kabanda, A. (2024). *Listening to distant oceans reveals subsurface temperatures in the Western Branch of the East African Rift System (EARS), Uganda*. Lightning talk at the Northwestern CoDEX Symposium, Northwestern University, April 2, 2024.

2023

Kabanda, A., Alonzo, B., van der Lee, S., Stamps, D. S., Atekwana, E. A., Nyago, J., Kabenge, L., Tugume, F., Fishwick, S., Kolawole, F., Evans, R. L., Taylor, M. H., Katumwehe, A. B., Atekwana, E. A., & Kiberu, J. M. (2023). *Interferometry of ambient seismic noise recorded by DRIAR stations in the*

northern Western Branch of the East African Rift System, Uganda. Poster T13C-0234, AGU Fall Meeting, San Francisco, California, December 11–15, 2023. **Outstanding Student Presentation Award (OSPA)**.

Kabanda, A., Alonzo, B., van der Lee, S., Stamps, D. S., Atekwana, E. A., Nyago, J., Kabenge, L., Tugume, F., Fishwick, S., Kolawole, F., Evans, R. L., Taylor, M. H., Katumwehe, A. B., Atekwana, E. A., & Kiberu, J. M. (2023). *Characteristics of ambient seismic noise recorded by DRIAR stations along the Western Branch of the East African Rift System, Uganda*. Presentation at the Southern East African Rift System Workshop, Blantyre, Malawi, July 25–28, 2023.

2022

AGU Fall Meeting 2022 – Attendee during the first year of PhD research.

SUPERVISION

- Isabela M, Undergraduate summer student intern, Carleton College, (2026)
- Joann M, Undergraduate summer student intern, Carleton College, (2026)

Isabela and Joann both worked on a seismology project of data analysis and crustal structure of Greenland under my supervision and guidance. They analyzed both directional and seasonal variations in seismic data, measured group and phase velocities from the extracted surface waves, computed velocity changes, and compared all these with independent datasets e.g., significant wave height, snowfall rate, rain precipitation, and others.

- Faith Momoh, NU-Geopaths, High school summer intern, 2023

Faith worked on a seismology project about seismometer construction using cardboards and also research about earthquakes in her home country, how many, where, and why there.

AWARDS, LEADERSHIP, AND MEMBERSHIPS

- SSA Student Travel Grant, 2026.
- Northwestern University Conference and Travel Grant, 2026.
- AGU Outstanding Student Presentation Award, 2023.
- AGU Seismology Student Representative and Executive Committee member, 2024–2027; organize student seismology workshops and professional-development activities.
- Member, American Geophysical Union, 2022–present.
- Member, Seismological Society of America, 2022–present.